

Effect of automation transparency in the management of multiple unmanned vehicles

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ABSTRACT

Automated decision aids can undoubtedly benefit system performance, but have the potential to provide incorrect advice, creating the possibility for automation disuse or misuse. This paper examined the extent to which increased automation transparency could improve the accuracy of automation use in a simulation of unmanned vehicle control. Participants were required to assign the best unmanned vehicle to complete missions. An automated recommender system provided advice but was not always reliable. Three levels of automation transparency were manipulated between-participants. Providing transparency regarding the reasoning underlying automated recommendations improved the accuracy of automation use, with no cost to decision time or subjective workload. Higher transparency in the form of projected decision outcome visualizations led to faster decisions, but less accurate automation use and an automation bias. Implications for the design of transparent interfaces to improve human-autonomy teaming outcomes in time-pressured environments are discussed.

1. Introduction

Automated decision aids are used in a range of industrial contexts to reduce human cognitive workload and extend work system capability (Sheridan, 2015; Vagia et al., 2016). For example, in healthcare, automated decision aids integrate information to create patient diagnoses and treatment recommendations (Lai et al., 2016); in air traffic control, separation assurance decision aids give recommendations to controllers regarding how to keep aircraft safely separated (Trapsilawati et al., 2016); in driving, automated systems provide support to drivers by deciding when it is safe to perform a lane change (Banks and Stanton, 2016); in airport baggage screening, decision support systems help operators detect prohibited items (Huegli et al., 2020); and in railway control, decision aids advise operators how to route trains (Balfe et al., 2015). Further, in the defense and security sectors, automated decision aids with complex computational algorithms have been developed to recommend the deployment of unmanned vehicles (UVs) for missions such as surveillance, delivering cargo, and search and rescue (Calhoun

et al., 2018; Endsley, 2017; U.S. Airforce, 2015).

While automation undoubtedly benefits system performance, contextual factors in complex work environments can mean that automation has the potential to provide incorrect advice. This uncertainty creates the possibility for the human operator to disuse or misuse automation (Lee and See, 2004; Parasuraman; Manzey, 2010; Parasuraman and Riley, 1997). Disuse refers to operators not actioning correct automated recommendations, whereas misuse refers to operators actioning incorrect automated recommendations. Another issue is that due to the complexity of automation, operators may need significant time to verify recommendations, potentially increasing decision time or mental workload.

Providing feedback about automation reliability, or an explanation for automation errors, can facilitate appropriate trust and improve performance in detection and identification tasks (e.g., Bagheri and Jamieson, 2004; Dzindolet et al., 2003; Seong and Bisantz, 2008; Wang et al., 2009). Related to this, increasing the *transparency* of automation (also referred to in some literature as automation *visibility* or automation

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observability; e.g., Dorneich et al., 2017; Woods, 1996) has been identified as a work design intervention that may allow operators to more accurately and efficiently use automation by facilitating their understanding of the reasoning underlying automated recommendations and what they should expect to happen if they were followed (Chen and Barnes, 2014; Hoff and Bashir, 2015; Lyons et al., 2017; Matthews et al., 2020). The key question raised in the contemporary automation transparency literature has concerned how to design transparent automation which allows operators to appropriately accept or reject recommendations, without increasing decision time or workload.

A model of transparency called the Situation awareness-based Agent Transparency (SAT) has been proposed by Chen et al. (2014), based on Endsley's (1995) three-level concept of situation awareness formation and Lee and See's (2004) theory of human trust in automation (see also Robot-to-Human Transparency model by Lyons, 2013). The SAT model provides guidance on what information should be communicated to operators to promote automation transparency. The SAT model proposes that Level 1 transparency should provide the operator with information about the automation purpose/intent (*What is the automation trying to achieve?*), Level 2 transparency provides information about the rationale behind the decision advice (*Why does the automation recommend this action?*), and Level 3 transparency provides information regarding the projection of future outcomes and uncertainty (if any) associated with those projections (*What does the automation expect to happen if I follow this recommendation?*). Relevant information across all three levels allows the operator to recognize when the automation can be relied upon, though Chen et al. (2014) note that all three levels may not be necessary to obtain the positive effects of transparency, depending on the task context.

In a relatively small number of studies, the SAT model has been used as the primary overarching theoretical framework to operationalize automation transparency and examine the impact of increased transparency on the accuracy of automation use (e.g., Mercado et al., 2016; Selkowitz et al., 2016; Stowers et al., 2020; Wright et al., 2015, 2017). A variety of task paradigms and outcome measures have been used which make direct comparison between studies difficult (see Bhaskara et al., 2020, for a review). For example, some studies have provided operators with automated recommendations regarding how to direct ground vehicles or dismounted soldiers around obstacles to their destination (Selkowitz et al., 2016; Wright et al., 2015, 2017, 2020). Some, but not all, of these studies have found that operators can use automation more accurately with increased transparency, however this can come at the cost of increased decision time (e.g., Wright et al., 2015).

Our current study is more closely related to the work of Mercado et al. (2016) and Stowers et al. (2020), where the participants' task was to select the most appropriate UV to complete missions, by either following a recommended best UV (Plan A) or rejecting it by choosing the recommendation for the second-best UV (Plan B). On the majority of trials (62.5%) in the Mercado et al. study, Plan A was optimal but on some trials (37.5%) Plan B was optimal due to a change in mission priorities. At Level 1 transparency, participants were only provided with basic plan information—what Plans A and B were and the routes these two UVs would take. At Level 1+2 transparency, participants were also provided with the rationale behind plans in graphical and text formats: a pie chart with different sizes of the pie slice (larger = more important) and color (green = high vs. yellow = suboptimal) illustrating the importance and suitability of the UV capabilities associated with Plans A and B, and a rationale conveyed in a text format (e.g., "Adequate coverage will be achieved by [vehicle] A1"). At Level 1+2+3 transparency, in Stowers et al., participants additionally received information in the graphic and text box indicating the relative projected outcomes of Plans A and B (e.g., circles representing UV characteristics were placed on sliding scales so that the projected success of each plan could be compared). At Level 1+2+3 transparency, in Mercado et al., participants received information about the uncertainty of the plans, which was displayed through the opacity of the graphics/icons and conveyed

in text. The accuracy of automation use generally improved with increased transparency (Mercado et al.; Stowers et al.).

Given the relative infancy of using the SAT model as a theoretical framework to guide the design of automation transparency, it is critical to continue to establish the generalizability and boundary conditions of the SAT model design principles by examining the impact of automation transparency under various task conditions. Operators of an unmanned vehicle system are typically required to monitor and process quantitative data at regular intervals (such as those related to UV characteristics of fuel and speed). Our UV task paradigm drew upon several concepts from Mercado et al. (2016) and Stowers et al. (2020), however, it significantly differs in that we were particularly interested in UV contexts in which operators need to make decisions under time pressure on complex quantitative data (as opposed to the qualitative reasoning required in the Mercado et al. and Stowers et al. studies). Under these contexts, the automated decision aid supports the computation and integration of data from the environment, which can be quite a cognitively challenging task for human operators to do themselves in the time available.

1.1. Experimental paradigm and predictions

Participants in the current study selected the most appropriate UV to complete missions, and received assistance from automation (the Recommender) that provided two plans: Plan A (best UV) and Plan B (second-best UV). The Recommender derived the plans by evaluating each UV's capabilities against three mission attributes: Time to Search Area, Search Time, and Fuel Consumption (the relative importance of the three mission attributes differed on each trial). Participants were told that the Recommender was highly reliable but would not always be accurate because it was not privy to Intel, which could change the optimality of the plans. The task was designed such that in 25% of cases, Plan B should be selected but participants were not aware of the actual reliability rate. On each trial, participants were presented four UVs at various locations. Then they were presented a mission, at which time only two of the four original UVs remained on the tactical map (recommended as Plans A and B). Participants evaluated the relative importance of the mission attributes, the two UV's capabilities (based on the UV's data such as fuel and speed) against these attributes, and the impact of Intel to decide between Plans A and B.

Our manipulation of automation transparency was based on the SAT model (Chen et al., 2014), and to facilitate comparison to prior work, we adopted the SAT notation of Level 1, Level 1+2, and Level 1+2+3 transparency. Participants in the Level 1 transparency condition were informed that the Recommender would use the UV's data to evaluate the UV's capabilities against the weighted mission attributes (the relative weighting of mission attributes was shown to participants) to determine Plans A and B. Participants in the Level 1+2 transparency condition were additionally presented with the actual formulae that the Recommender used to combine the UV's data to calculate their capabilities on the weighted mission attributes, thereby communicating to the participant the underlying analytical structure of the recommendation (i.e., of how the automation is doing the analysis and how it makes decisions) in a manner consistent with the SAT model. Participants in the Level 1+2+3 transparency condition were additionally provided with a visualization of the projected outcomes of Plans A and B. The projected outcomes displayed how the two UVs chosen for Plans A and B would perform relative to each other on each of the weighted mission attributes (i.e., following the SAT model; this conveyed to the participant what the automation expected to happen if the recommendation was followed). Due to the complexity of our UV task, we did not include environmental uncertainty and thus no associated representation of uncertainty was included in our Level 1+2+3 transparency design.

On this basis of the SAT model, it was predicted that increased transparency would assist participants to accurately assess the impact of Intel on the recommended plans. To test this we compared the accuracy

of automation use between the Level 1+2 and Level 1 transparency conditions, and in turn between the Level 1+2+3 and Level 1+2 transparency conditions. We predicted that with increased transparency, participants would use automation more accurately without impairing decision time. In our task paradigm, we did not expect participants to have the time to verify recommendations by making precise manual calculations; nonetheless, we expected that the provision of the formulae containing the Recommender's reasoning of how it came up with Plans A and B (Level 1+2 transparency) would be more useful for assessing the Recommender data computations and the impact of Intel than only being provided weighted mission attributes (Level 1 transparency). In turn, being provided the projected outcome visualizations (Level 1+2+3 transparency)—that showed how the UVs for Plans A and B would perform relative to each other on the weighted mission attributes—was predicted to be more useful for assessing the Recommender data computations and the impact of Intel compared to Level 1+2 transparency (where weighted attributes and UV capability formulae were provided, but with no integration of the two). To be clear, although the projected outcomes could also be made through calculation based on the information available at Level 1+2 transparency, Level 1+2+3 transparency actually provided and *displayed* the resultant projected outcomes, thereby providing additional diagnostic information for assessing the impact of Intel. Our study also included subjective measures of workload, automation trust, and automation usability: There has been no evidence from prior studies to suggest that increased transparency negatively impacts workload, and the impact of transparency on these latter two outcomes have been mixed (see Bhaskara et al., 2020, for a review).

2. Method

2.1. Participants

One-hundred and seventy-six undergraduate students (56% female; $M = 21.92$ years; $SD = 6.45$ years) at the University of Western Australia (UWA) participated in return for course credit. Participants were randomly assigned to either Level 1 ($n = 60$), Level 1+2 ($n = 59$), or Level 1+2+3 ($n = 57$) transparency condition. This research was approved by the Human Research Ethics Office at UWA. Informed consent was obtained from each participant.

2.2. Unmanned vehicle task

The UV task was presented through two displays (Fig. 1). The mission window (left display) included panels for mission information. The tactical map (right display) included a side panel for UV data. Participants were required to deploy the optimal UV to complete each mission. There were 48 unique missions created across three terrains (urban, rural, and ocean). Each trial started with a 30-s familiarization phase followed by a 30-s mission phase (one mission was presented per trial).

2.2.1. Familiarization phase

Participants prepared for each mission by familiarizing themselves with the four UVs and the Intel (Fig. 2). The UVs were numbered 1 to 4, and could either be aerial (UAV), ground (UGV), or surface (USV) vehicles. The tactical map showed the UV's call-signs (e.g., UAV3) and locations. The UV's speed and fuel efficiency data were displayed on the side panel. Three Intel messages appeared within the Intel panel at 9-, 12-, and 27-s into familiarization. Participants were required to click on the Intel panel to open and read the Intel. Each familiarization phase lasted for 30-s.

To assess workload during the familiarization phase, participants completed a secondary task. An audio track played a series of radio signal combinations. Each combination consisted of one call-sign (Alpha, Bravo, Charlie, Delta, or Echo), one color (Blue, Green, Red,

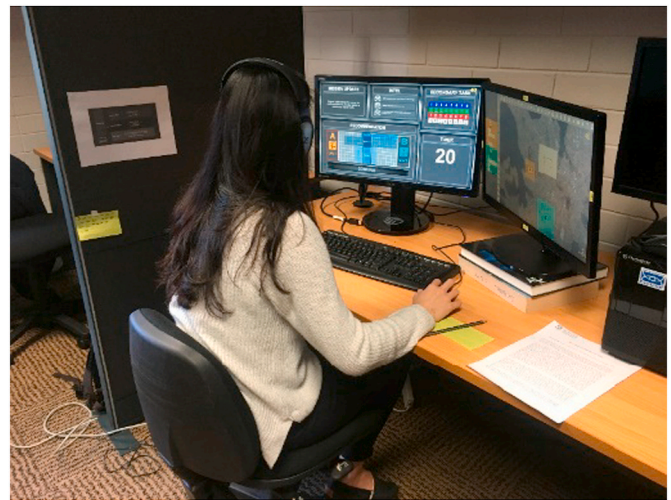


Fig. 1. The UV task presented through the mission window (left display) and the tactical map (right display). A verbal definition of each mission attribute (e.g., Time to Search Area = How much time it takes the UV to reach the search area) was visible on the wall to the left for all participants. The Recommender's actual formulae (e.g., Time to Search Area = UV's distance to search area/speed) was also visible on the wall to the left, but only for the Level 1+2 and Level 1+2+3 transparency conditions. Participants had a scratch pad and pencil to their right to use if they wanted to make any notes.

or White), and one number (1–8). On each trial participants listened to six call-sign color and number combinations (e.g., Alpha Blue 2, Charlie Red 8), with the first presented at 3-s, and the remaining at 5-s intervals. Participants were required to listen for Charlie (which occurred twice during each familiarization trial), and input the subsequent color and number (e.g., Red 8) using the Secondary Task panel in the Mission Window.

2.2.2. Mission phase

After the familiarization phase, a mission (e.g., “A suspicious boat is operating aimlessly around the Eastern Islands. Provide surveillance”) appeared in the Mission Update panel (Figs. 3 and 4). At mission onset, the mission location and associated search area appeared on the tactical map as a translucent grey box, and unopened Intel (if any) was automatically opened. As shown in Figs. 3 and 4, the Recommendation panel displayed a mission attribute bar (the single vertical bar) and the two recommendations (Plan A [USV4] and Plan B [UAV3]). Only the two recommended UVs remained on the tactical map, and two additional sources of data associated with the two UVs (distance-to-search-area and search-steps) were displayed on the side panel of the tactical map. The Timer panel counted down the 30-s available for participants to select, by clicking, either Plan A or B. Participants then clicked the “Confirm” button to deploy the UV. The secondary task was inactive during the mission phase.

The Recommender used the UV's data (speed, fuel efficiency, distance-to-search-area, and search-steps) to calculate each UV's capability on three mission attributes: (a) Time to Search Area (how much time it takes the UV to reach the search area); (b) Search Time (how much time the UV takes to search the area); and (c) Fuel Consumption (how much fuel the UV uses to complete the mission; see Figs. 1 and 5). These mission attributes had different relative importance weightings on each trial and these weightings were displayed in the mission attribute bar with larger proportions indicating a higher weighting (in Figs. 3 and 4, Time to Search Area had the highest weighting). The UV score on each mission attribute was multiplied by the weighting for that attribute, and

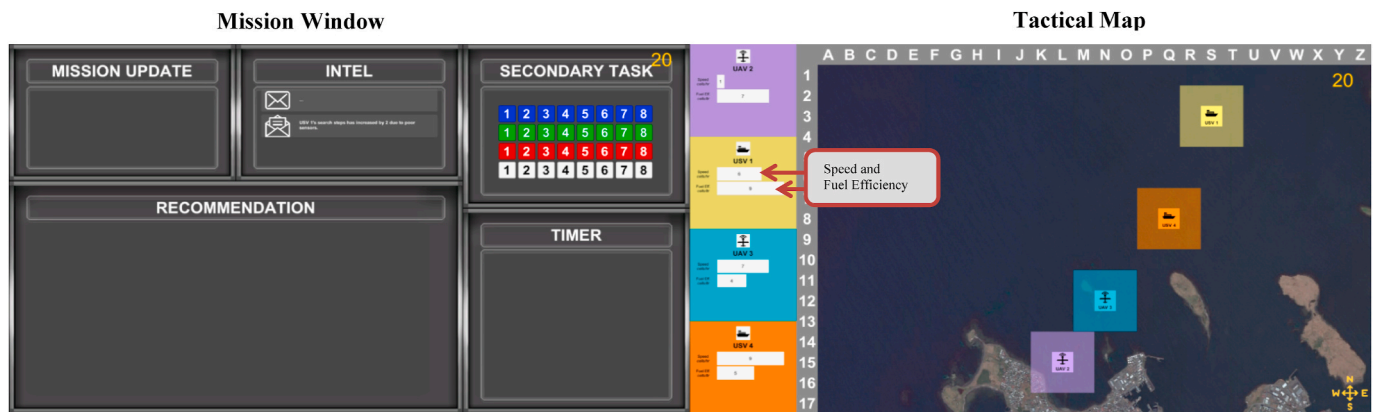


Fig. 2. The UV task displaying the mission window (left) and the tactical map (right) during the familiarization phase. The mission window contained the Mission Update, Intel, Recommendation, Secondary Task, and Timer panels. The tactical map showed the four UVs with their speed and fuel efficiency data (displayed numerically and visually through the horizontal white bars) on the side panel. The ocean terrain map was used in this example. The display shown for the familiarization phase was identical across the three transparency conditions.



Fig. 3. The UV task displaying the mission window (left) and the tactical map (right) during the mission phase (in the Level 1 and Level 1+2 transparency conditions). The Mission Update and Intel panels showed the mission and intelligence information, respectively. In the Recommendation panel, Plans A and B showed the two most optimal UVs and the mission attribute bar in the middle depicted the relative importance weightings of the mission attributes (i.e., Time to Search Area [the most important attribute for this specific mission], Search Time, and Fuel Consumption). The Timer panel showed the time remaining to make a decision which counted down from 30-s. The tactical map showed the two recommended UVs and two additional sources of data for the two UVs (distance-to-search-area and search-steps; displayed numerically and visually through the horizontal bars) on the side panel. The translucent grey box depicted the mission location and its associated search area. Participants selected, by clicking, either Plan A or B; and then clicked the Confirm button to deploy the UV.

then summed. The UV with the lowest overall score was the optimal vehicle for the mission and was selected by the Recommender as Plan A, while the UV with the next lowest overall score was selected as Plan B.

Plan A was the optimal plan on 75% of trials. In 25% of trials, Plan B was the optimal plan due to Intel that the Recommender could not access. Three Intel messages (all text-based) were provided on each trial. Intel could either have a positive impact on a UV property (e.g., Intel 1 in Fig. 3 made the UV closer to the search area, “Updated GPS coordinates show that UAV3 is 3 cells due East”); a negative impact (e.g., “USV 1’s search steps has increased by 2 due to poor sensors”); or no direct impact (e.g., “Weather forecast for tomorrow is calm”). Participants were told that only relevant Intel that explicitly stated a quantitative change in the data or position of the UV associated with either Plan A or B (e.g., “Updated GPS coordinates show that UAV3 is 3 cells due East”) could potentially impact the accuracy of recommendations (relevant Intel was present in 30 of the 36 test trials [83%]). Intel impacting either of the two non-selected UVs, or other Intel (e.g., “USV 1’s search steps has increased by 2 due to poor sensors,” or “Weather forecast for tomorrow is calm”), could not impact recommendations.

In nine of these 30 test trials (30%), the relevant Intel was of sufficient magnitude to make Plan B the optimal plan. Thus, Plan B was not always the optimal plan on trials with relevant Intel (only 30% of the time), as this depended on the relative direction and magnitude of the impact of that Intel on Plans A and B as a function of the UV’s data and the mission attributes weightings. When the optimal plan changed, it was due to one of three Intel messages, and not some combination of more than one relevant Intel. A full list of the mission descriptions and Intel messages is provided in the Appendix. The training trials consisted of unique missions that were different from the test trials and had the same ratio of Plan A being optimal (on 75% of trials) as the test trials, and a similar ratio of trials with relevant Intel.

2.2.3. Manipulation of transparency

In the Level 1 transparency condition, participants were instructed that the Recommender would evaluate each UV’s capabilities against the weighted mission attributes (the weighted mission attributes were shown to participants on the mission window; see Fig. 3) to determine Plans A and B. A copy of the verbal definition of each mission attribute



Fig. 4. The UV task displaying the mission window (left) and the tactical map (right) during the mission phase (in the Level 1+2+3 transparency condition). The Mission Update and Intel panels showed the mission and intelligence information, respectively. In the Recommendation panel, Plans A and B showed the two most optimal UVs and the mission attribute bar in the middle depicted the relative importance weightings of the mission attributes (i.e., Time to Search Area [the most important attribute for this specific mission], Search Time, and Fuel Consumption). In the Level 1+2+3 transparency condition, the additional visualization of projected outcomes of the two plans was also presented via the horizontal blue-shaded bars. These projected outcomes displayed how the two UVs chosen for Plans A and B would perform (relative to each other) on each of the weighted mission attributes. The Timer panel showed the time remaining to make a decision which counted down from 30-s. The tactical map showed the two recommended UVs and two additional sources of data for the two UVs (distance-to-search-area and search-steps; displayed numerically and visually through the horizontal bars) on the side panel. The translucent grey box depicted the mission location and its associated search area. Participants selected, by clicking, either Plan A or B; and then clicked the Confirm button to deploy the UV. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

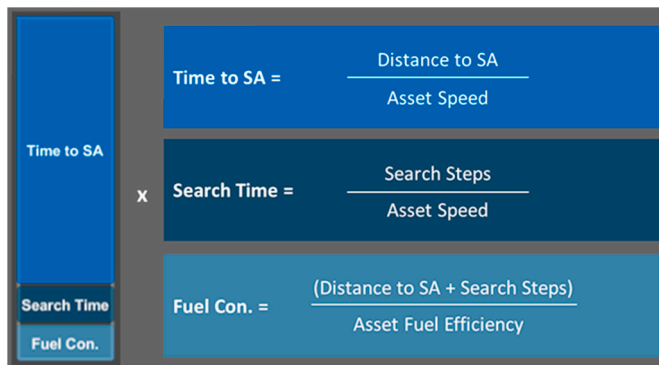


Fig. 5. The formulae that the Recommender used based on UV's data to calculate UV's capabilities (right) against the weighted mission attributes (left) to determine Plans A and B. The relative weighting of the mission attributes were mission-specific and thus differed on each trial. Participants in the Level 1+2 and Level 1+2+3 transparency conditions were provided a hard copy of Fig. 5.

(Time to search area = how much time it takes the UV to reach the search area; Search Time = how much time the UV takes to search the area; Fuel Consumption = how much fuel the UV uses to complete the mission) was provided and located on the wall to their left (see Fig. 1). Participants in the Level 1 transparency condition were not presented a hard copy of the recommender formulae.

In the Level 1+2 transparency condition, the mission window and the tactical map displays were identical to the Level 1 transparency condition (Fig. 3), but participants were additionally informed of the Recommender's formulae (see Fig. 5) and had a hard copy of these formulae visible (along with the verbal definition of each mission attribute as shown in Level 1 transparency) on the wall to the left of them (Fig. 1).

In the Level 1+2+3 transparency condition, in addition to the above definitions and formulae, participants received a visualization of the relative capability projection of the UVs associated with Plans A and B

(Fig. 4). The projected outcomes depicted how each UV chosen for each plan would perform relative to each other (as indicated by the horizontal blue-shaded bars on each of the weighted mission attributes), which could then be used to assess the impact of Intel on the recommended plans. The UV with the lowest score was selected as the optimal plan by the Recommender; depicted in Fig. 4, the horizontal blue-shaded bars for USV 4 (Plan A) occupied less space overall than UAV3 (Plan B). Specifically, USV4 has shorter bars for Time to Search Area and Fuel Consumption, indicating that USV4 would get to the search area much quicker and consume slightly less fuel than UAV3. However, due to Intel 1 ("Updated GPS coordinates show that UAV3 is 3 cells due East"), UAV3's improvement in distance-to-search-area was sufficient to make Plan B better than Plan A: because UAV3's new position has now made its Time to Search Area faster than USV4, and Time to Search Area was a heavily weighted mission attribute. The projected outcome visualization bar graphs did not change as a function of Intel because the Recommender was unaware of the Intel, rather the participant needed to assess the impact of Intel on the recommended plans. Participants in the Level 1+2+3 transparency condition were trained on the relationship between potential Intel messages and the corresponding changes to expect in the projected outcome visualization. Given time pressure, participants would not necessarily have the time to verify recommendations by making precise calculations (e.g., calculating UAV3's exact amended Time to Search Area; Fig. 4). Nonetheless, showing how the plans would perform relative to each other on each of the weighted mission attributes was expected to be more useful for estimating the impact of Intel compared to Level 1+2 transparency (where weighted attributes and formulae were provided, but with no integration of the two).

2.3. Measures

2.3.1. Accuracy of automation use

The Hit rate was calculated as the proportion of trials that the participant selected Plan A when Plan A was correct. The Correct Rejection rate was calculated as the proportion of trials that the participant selected Plan B when Plan B was correct. We used a signal detection analysis to assess the overall accuracy of automation use (Green and Swets, 1966). The parameter d' assessed the sensitivity of participants to correctly discriminate Plan A from Plan B (larger values

indicate better sensitivity). The parameter c assessed the degree to which participants were biased toward agreeing with the automation (choosing Plan A). Negative values of c signify a bias toward agreeing with the automation, whereas positive values signify a bias toward choosing Plan B. To adjust for extreme hit (0 or 1) and false alarm (0 or 1) values, we replaced rates of 0 with $0.5/n$ and rates of 1 with $(n - 0.5)/n$ where n is the number of signal (Plan A correct) or noise (Plan B correct) trials (Macmillan and Kaplan, 1985).

2.3.2. Decision time

Correct decision time was defined as the time taken to correctly select Plan A or B from the time of mission onset. Incorrect decision time was defined as the time taken to incorrectly select Plan A or B from the time of mission onset.

2.3.3. Subjective workload

The NASA task load index questionnaire (NASA-TLX) required participants to rate their workload, ranging from 0 (*low*) to 100 (*high*), on six dimensions of workload: mental demands, physical demands, temporal demands, own performance, effort, and frustration. Participants then indicated the degree to which each of these six dimensions was “the more important contributor to workload” in pairwise comparisons between the dimensions (Hart and Staveland, 1988). The overall NASA TLX workload score was calculated by multiplying the resulting weighting of each dimension with the corresponding rating, then dividing the total by the number of pairwise comparisons.

2.3.4. Secondary task performance

Accuracy on the secondary task was calculated as the proportion of correct responses to the 36 presented call-sign, color, and number combinations (i.e., correct by responding when the call-sign Charlie was presented, or correct by not responding when Charlie was not presented).

2.3.5. Trust questionnaire

Trust in the Recommender was assessed using a modified version of the Merritt (2011) scale. The scale included six items such as, “I believe Recommender is a competent performer” and “I trust Recommender.” Items were measured on a 5-point likert response scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*).

2.3.6. System usability scale (SUS)

Perceived usability was measured using the SUS (Brooke, 1996), a ten item survey on a 5-point likert scale, scoring from 0 (*strongly disagree*) to 4 (*strongly agree*). Example questions include, “I think that I would like to use the Recommender frequently” and “I found the Recommender unnecessarily complex.” The average trust in automation score was calculated by averaging responses to the 5 items. For the SUS, some items were first reversed scored, and then the combined score was multiplied by 2.5 to yield an SUS score from 0 to 100.

2.4. Procedure

Participants were first provided with 35-min condition-specific (Level 1, Level 1+2, or Level 1+2+3 transparency) comprehensive audio-visual instructions, followed by a video demonstration. These instructions contained an explanation of the UV task as presented above, their role as the operator, verbal definition of the UV data and mission attribute variables, and the trial structure—all participants were also provided with an example of a mission, how to interpret the mission attributes weightings, and a reminder that the Recommender was not privy to Intel. For the Level 1+2 and Level 1+2+3 transparency conditions, participants were additionally explained how the Recommender applied the formulae (the numerical calculations of the UV's scores based on UV's data against the weighted mission attributes to determine Plans A and B). For the Level 1+2+3 transparency condition,

participants were additionally shown the visualization of these calculations (projected outcomes of Plans A and B) and instructed how to interpret the visualization bar graphs together with the mission attributes weightings. In line with prior research (e.g., Mercado et al., 2016), participants were informed that while the Recommender was highly reliable, it would not be 100% accurate. Participants then completed 12 training trials with feedback, during which the reliability of the automation was 75% (i.e., in 3 of the 12 trials, Plan B was optimal). This was followed by 36 test trials without feedback, during which the reliability of the automation was 75% (i.e., in 9 of the 36 trials, Plan B was optimal). There was a 60-s break after 18 test trials. At the end of the experiment, participants completed the NASA-TLX twice: first for their recalled workload during familiarization phases, and second for their recalled workload during mission phases. Participants then completed the trust and SUS scales in a counterbalanced order. The experiment duration was 2 h.

3. Results

Participants on average responded to 0.80 ($SD = 0.10$) of the training missions correctly. A 3 (Level 1, Level 1+2, Level 1+2+3 transparency) $\times$ 2 (first-half, second-half training block) mixed Analysis of Variance (ANOVA) revealed a main effect of training, $F(1, 173) = 5.88, p = .02, \eta_p^2 = 0.03$, with higher accuracy during the second-half of training ($M = 0.82, SD = 0.15$) compared to the first-half of training ($M = 0.78, SD = 0.18$). There was no main effect of condition, $F(2, 173) = 1.48, p = .23$, and no interaction, $F(2, 173) = 1.38, p = .25$.

Table 1 presents the overall descriptive statistics and inter-correlations between the dependent variables. The accuracy with which participants used automation (parameter d' , which assessed the sensitivity of participants to correctly discriminate Plan A from Plan B) was strongly positively associated with the proportion of trials that participants selected Plan A when Plan A was correct (Hit rate), and the proportion of trials that the participants selected Plan B when Plan B was correct (Correct Rejection rate). As would be expected, the Hit rate was strongly associated with a response bias (c) toward agreeing with automation (choosing Plan A), while the Correct Rejection rate was strongly associated with a bias toward choosing Plan B. As would be expected, strong positive associations were found between decision time for correct and incorrect decisions, and between subjective workload ratings for the familiarization and mission phases. There was also a strong positive association between trust in automation and the degree to which participants rated the automation as useable. Several other weak-to-moderate correlations were also found. For example, those participants who more often selected Plan B when Plan B was correct (Correct Rejection rate) took longer to make both correct and incorrect decisions, and those participants who used automation more accurately (parameter d') rated the automation as more useable.

Table 2 presents the descriptive statistics for the test trials for each dependent variable as a function of transparency.

3.1. Data analysis and statistics

We analyzed Hit rate, Correct Rejection rate, signal detection, trust, secondary task performance, and usability data using one-way ANOVAs. We followed up any significant one-way ANOVAs with planned contrasts (t -tests) that directly paralleled our hypotheses by comparing the Level 1+2 to Level 1 transparency condition, and the Level 1+2+3 to Level 1+2 transparency condition (Rosenthal and Rosnow, 1985).

We analyzed decision time using a 3×2 mixed-design ANOVA with a between-participants factor of transparency condition (Level 1, Level 1+2, Level 1+2+3) and a within-participants factor of decision outcome (correct, incorrect). We then followed up any significant main effect of condition and interaction with simple effect tests (one-way ANOVAs) comparing the conditions at each level of decision outcome. The same methods were used to analyze workload data, whereby a 3×2

Table 1

Overall means, standard deviations, and intercorrelations between accuracy of automation use (hit, correct rejection, sensitivity, and response bias), decision time, subjective workload, trust in automation, and perceived usability of automation.

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10
1. Hit rate	0.84	0.12	–									
2. Correct rejection rate	0.71	0.18	-.07	–								
3. Sensitivity (<i>d'</i>)	1.73	0.73	.61**	.72**	–							
4. Response bias (<i>c</i>)	–0.23	0.39	-.66**	.77**	.13	–						
5. Decision time (correct)	14.45	4.11	-.00	.20**	.12	.17*	–					
6. Decision time (incorrect)	17.92	5.55	.12	.23**	.21**	.12	.80**	–				
7. Workload (familiarization phase)	49.92	19.55	.11	-.24**	-.14	-.22**	.05	.03	–			
8. Workload (mission phase)	69.23	17.84	.02	-.01	-.00	-.01	.29**	.26**	.47**	–		
9. Trust	3.69	0.62	.09	-.04	.07	-.11	-.09	-.01	-.05	.09	–	
10. Usability	73.93	13.46	.20**	.03	.19*	-.12	-.16*	-.08	-.17*	-.10	.60**	–

Note. Decision time was measured in seconds. Maximum scores for workload and usability = 100. Maximum score for trust = 5. Weak, moderate, and strong effect sizes are 0.10, 0.30, and 0.50, respectively (Cohen, 1992).

* $p < .05$. ** $p < .01$.

Table 2

Mean accuracy of automation use (hit, correct rejection, sensitivity, and response bias), decision time, subjective workload, trust in automation, and perceived usability of automation as a function of transparency condition.

	Transparency		
	Level 1	Level 1+2	Level 1+2+3
Hit rate	0.80 (0.13)	0.85 (0.11)	0.86 (0.11)
Correct rejection rate	0.73 (0.19)	0.76 (0.15)	0.65 (0.19)
Sensitivity (<i>d'</i>)	1.61 (0.74)	1.93 (0.69)	1.65 (0.73)
Response bias (<i>c</i>)	–0.12 (0.39)	–0.19 (0.34)	–0.38 (0.40)
Decision time (seconds)			
Correct decision	14.54 (4.18)	15.66 (3.74)	13.17 (4.08)
Incorrect decision	17.29 (5.86)	19.56 (4.86)	16.90 (5.60)
Subjective workload			
Familiarization phase	49.89 (19.15)	48.03 (21.01)	51.81 (18.57)
Mission phase	72.09 (17.75)	69.98 (15.48)	65.78 (19.72)
Trust	3.60 (0.66)	3.75 (0.57)	3.71 (0.63)
Usability	73.60 (13.84)	75.00 (14.87)	73.21 (11.68)

Note. Standard deviations are in parenthesis.

(Condition [Level 1, Level 1+2, Level 1+2+3] $\times$ Task Phase [familiarization, mission]) mixed ANOVA was conducted followed by simple effect tests (one-way ANOVAs) comparing the conditions at each level of task phase. For both the decision time and workload data, we followed up any significant one-way ANOVAs with planned contrasts (*t*-tests) that directly paralleled our hypotheses by comparing the Level 1+2 to Level 1 transparency condition, and the Level 1+2+3 to Level 1+2 transparency condition.

We estimated effect sizes for *F*-tests using partial eta squared (small = 0.01, medium = 0.06, large = 0.14), and for *t*-tests using Cohen's *d* (small = 0.20, medium = 0.50, large = 0.80; Cohen, 1992).

3.2. Accuracy of automation use

There was a significant difference in Hit rate between the conditions, $F(2, 175) = 5.12, p = .007, \eta_p^2 = 0.06$. The Hit rate was higher for participants in the Level 1+2 transparency condition compared to participants in the Level 1 transparency condition, $t(114) = 2.40, p = .018, d = 0.41$, but there was no difference in Hit rate between the Level 1+2+3 and Level 1+2 transparency conditions, $t(117) = 0.55, p = .58$.

There was a significant difference in Correct Rejection rate between the conditions, $F(2, 175) = 5.58, p = .004, \eta_p^2 = 0.06$. The Correct Rejection rate did not differ between participants in the Level 1+2 and Level 1 transparency conditions, $t(114) = 1.01, p = .31$, but the Correct Rejection rate was lower for participants in the Level 1+2+3 transparency condition compared to participants in the Level 1+2 transparency condition, $t(117) = 3.36, p < .001, d = 0.64$.

There was a significant difference in sensitivity (*d'*) between the conditions, $F(2, 175) = 3.31, p = .039, \eta_p^2 = 0.04$. As expected,

participants in the Level 1+2 transparency condition used automation more accurately than participants in the Level 1 transparency condition, $t(114) = 2.36, p = .02, d = 0.44$. However, contrary to our predictions, participants in the Level 1+2+3 transparency condition used automation less accurately than participants in the Level 1+2 transparency condition, $t(117) = 2.10, p = .037, d = 0.38$.

As participants were told that the automation was reliable, it is not surprising that they showed a bias toward agreeing with the automation (choosing Plan A). However, there was a significant difference in response bias (*c*) between the conditions, $F(2, 175) = 7.21, p < .001, \eta_p^2 = 0.08$. We found no difference in response bias between participants in the Level 1+2 and Level 1 transparency conditions, $t(114) = 0.99, p = .33$. Participants in the Level 1+2+3 transparency condition were more biased toward agreeing with automation than participants in the Level 1+2 transparency condition, $t(117) = 2.76, p = .007, d = 0.51$.

3.3. Correct and incorrect decision time

A 3×2 (Condition [Level 1, Level 1+2, Level 1+2+3] $\times$ Decision Outcome [correct, incorrect]) mixed ANOVA was conducted on decision time. We found a significant main effect of decision outcome, $F(1, 173) = 193.75, p < .001, \eta_p^2 = 0.53$, indicating that participants took longer to make incorrect ($M = 17.92$ s, $SD = 5.44$ s) compared to correct decisions ($M = 14.45$ s, $SD = 4.11$ s). We found a significant main effect of condition, $F(1, 173) = 5.03, p = .008, \eta_p^2 = 0.06$, and no interaction between decision outcome and condition, $F(1, 173) = 2.07, p = .13$.

Follow up simple effect tests indicated a significant difference in correct decision time between the conditions, $F(2, 175) = 5.74, p = .004, \eta_p^2 = 0.06$. There was no difference in correct decision time between participants in the Level 1+2 and Level 1 transparency conditions, $t(114) = 1.51, p = .13$. However, participants in the Level 1+2+3 transparency condition made faster correct decisions than participants in the Level 1+2 transparency condition, $t(117) = 3.46, p < .001, d = 0.64$.

Follow up simple effect tests indicated a significant difference in incorrect decision time between the conditions, $F(2, 175) = 4.11, p = .018, \eta_p^2 = 0.05$. Participants in the Level 1 transparency condition made faster incorrect decisions than participants in the Level 1+2 transparency condition, $t(114) = 2.28, p = .02, d = 0.42$. Participants in the Level 1+2+3 transparency condition made faster incorrect decisions than participants in the Level 1+2 transparency condition, $t(117) = 2.77, p = .007, d = 0.51$.

3.4. Subjective workload and secondary task performance

A 3×2 (Condition [Level 1, Level 1+2, Level 1+2+3] $\times$ Task Phase [familiarization, mission]) mixed ANOVA was conducted on NASA-TLX subjective workload ratings. We found a significant main effect of task

phase, $F(1,173) = 181.74$, $p < .001$, $\eta_p^2 = 0.51$, indicating that participants recollected their workload as being higher during mission phases ($M = 69.23$, $SD = 17.84$) compared to during familiarization phases ($M = 49.92$, $SD = 19.55$). There was no main effect of condition, $F(1,173) = 0.37$, $p = .72$, but there was a significant interaction between condition and task phase, $F(1,173) = 3.57$, $p = .03$, $\eta_p^2 = 0.04$. However, follow up simple effect tests indicated no difference between the conditions in workload ratings for either the familiarization phase, $F(2,175) = 0.55$, $p = .87$, or the mission phase, $F(2, 175) = 1.92$, $p = .15$.

Accuracy on the secondary task during the familiarization phase was near ceiling ($M = 0.94$, $SD = 0.15$) and did not differ between the conditions, $F(2, 175) = 1.89$, $p = .15$.

3.5. Trust and perceived usability of automation

Trust in automation did not differ as a function of transparency condition, $F(2, 175) = 0.91$, $p = .404$. Ratings of automation usability also did not differ as a function of condition, $F(2, 175) = 0.28$, $p = .750$.

4. Discussion

The UV task in the current study was designed to emulate situations where operators need to make decisions under time pressure, and rely on automated algorithms to assist in computing and integrating complex quantitative data from the task environment. The operators' role was to accept or reject the recommendations made by automation, while assessing the impact of contextual factors (Intel) that only the operator could access. The aim was to examine the extent to which using the SAT model as a theoretical framework to guide the design of automation transparency, could improve human-autonomy teaming outcomes in this type of simulated UV task. The empirical findings can be summarized as follows. Participants in the Level 1+2 transparency condition used automation more accurately than participants in the Level 1 transparency condition. There was no difference in correct decision times between these conditions, but participants in the Level 1 transparency condition made faster incorrect decisions than those in the Level 1+2 transparency condition. Participants in the Level 1+2+3 transparency condition used automation less accurately, were more biased toward agreeing with the automation, and made faster correct and incorrect decisions than participants in the Level 1+2 transparency condition. There were no differences in ratings of workload, trust in automation, or perceived usability of automation as a function of transparency condition.

4.1. Level 1+2 transparency

Providing the rationale behind how the automation made recommendations (i.e., the formulae that the Recommender used to evaluate UV capabilities against the mission attributes) improved the accuracy of automation use. As noted earlier, comparison across studies needs to be done with caution because a variety of task paradigms and instantiations of presenting transparency have been used (see Bhaskara et al., 2020). Nonetheless, our finding of improved automation use with Level 1+2 compared to Level 1 transparency is consistent with prior research in ground vehicle convoy (Wright et al., 2017) and UV (Mercado et al., 2016) paradigms. Taken together, the outcomes from prior work and the current study indicate that the benefit of communicating the automation rationale, such as the underlying analytical structure of the automated decision (i.e., of how the automation is doing the analysis and how it makes decisions), appears to be reasonably robust across different task domains. We also demonstrated that Level 1+2 transparency can benefit automation use without increasing decision time or subjective workload (Mercado et al., 2016). While increased decision time is not necessarily a negative outcome (if it improves decision accuracy and system outcomes), it is advantageous if higher accuracy can be achieved without increased decision time. The fact that participants in the Level 1

compared to the Level 1+2 transparency condition made faster incorrect decisions suggests that they might have rushed their decisions because they did not have access to the underlying logic of the Recommender. The provision of Level 1+2 compared to Level 1 transparency did not improve automation trust or automation usability ratings (for similar results see Selkowitz et al., 2016; Wright et al., 2017; but also see Mercado et al., 2016).

4.2. Level 1+2+3 transparency

The finding that participants in the Level 1+2+3 compared to the Level 1+2 transparency condition used automation less accurately was not expected based on the SAT model. Specifically, providing participants with extra diagnostic information on how the UVs for Plans A and B would perform relative to each other on the weighted mission attributes did not allow participants to more accurately assess the impact of Intel on the recommended plans—it in fact had the opposite effect. Analysis of the Hit rate and Correct Rejection rate data, taken together with the signal detection analysis, indicated that participants in the Level 1+2+3 transparency condition used automation less accurately because they were significantly more biased toward agreeing with automation (choosing Plan A), leading to increased automation misuse (actioning incorrect recommendations, i.e., a lower Correct Rejection rate; Lee and See, 2004; Parasuraman and Manzey, 2010; Parasuraman and Riley, 1997). In line with this response bias, participants in the Level 1+2+3 transparency condition made faster correct and incorrect decisions than participants in the Level 1+2 transparency condition.

A similar response bias was found in a prior study using a clinical automated decision support system (Bussone et al., 2015). Participants provided with more comprehensive explanations about how the automated clinical system determined a diagnosis, showed a bias toward agreeing with the system (regardless of whether the system was accurate or not) and reported higher trust in the system, compared to participants provided with more selective explanations. The broader implication of the current study and that of Bussone et al. is that caution should be taken when considering the possible benefits of automation transparency, because providing more transparency information does not necessarily always lead to better outcomes. This conclusion is in line with the basic decision making literature which indicates that people have a general tendency to be more confident in their judgments when more information is presented, but that this is not necessarily accompanied by an increase in the accuracy of their judgments (e.g., Tsai et al., 2008).

It is possible that some form of *framing effect* might have also contributed to inducing the Level 1+2+3 transparency response bias (Plous, 1993). In the Level 1+2+3 transparency condition, the projected outcome visualization quite saliently highlighted (framed) the positive and negative outcomes associated with, respectively, Plans A and B. The heightened decision framing associated with the provision of the highly integrated Level 1+2+3 display information may have caused participants to more quickly evaluate each recommendation, resulting in faster decisions. That is, they might have been less willing to take risks to disagree with the automation (to reject Plan A), and therefore less likely to fully consider the impact of Intel on the recommended plans. As a result, when the Intel significantly affected one of the recommendations and Plan A was no longer optimal, there was an increased probability of making an erroneous decision (for a related discussion see Baber et al., 2019). In contrast, with Level 1+2 transparency, participants were not provided with such a strong visual frame of reference regarding the Recommender plans and likely had to rely more on their own estimates, resulting in longer but more accurate decisions.

4.3. Comparisons to Mercado et al. (2016) and Stowers et al. (2020)

Mercado et al. (2016) and Stowers et al. (2020) reported an increase in accuracy of automation use with the provision of Level 1+2+3

transparency compared to Level 1+2 transparency in UV tasks. More specifically, Mercado et al. reported an improved Correct Rejection rate for Level 1+2+3 transparency compared to Level 1+2 transparency, but no difference in Hit rate. Stowers et al. reported an improved Hit rate for Level 1+2+3 compared to Level 1+2 transparency, but no difference in the Correct Rejection rate. In contrast, we found an impaired Correct Rejection rate for Level 1+2+3 compared to Level 1+2 transparency, and no difference in the Hit rate. Our UV task paradigm shared similarities with the UV paradigms used in previous studies, but there are differences. The most notable difference concerns our inclusion of complex quantitative data and time pressure. In the Mercado et al. and Stowers et al. UV paradigm, the required reasoning for the recommended plans was qualitative (e.g., assessing the relative size/color of pie slices, opacity of graphics/icons, text-based reasoning). It may be more straightforward to implement Level 1+2+3 transparency in tasks that require qualitative decision-making, because the assessment is more “black-and-white” (you either have the extra diagnostic qualitative information or you do not). In contrast, our UV task required understanding of how the automation did the quantitative analysis, and although the visualization of projected outcome information did not increase subjective workload or decision time, it was not successful in allowing the operator to more accurately assess the impact of Intel. Instead, providing the projected outcome information caused participants to overestimate the reliability of the automation which led them to agree with the automation more; leading to faster but less accurate decisions.

4.4. Conclusions and outlook

Our results suggest that it may prove more difficult to design transparent automation that provides quantitative justifications of recommendations to operators that are under time pressure, without inadvertently causing them to overestimate the reliability of automated aids. The three levels of SAT-model transparency are designed to correspond to the three levels of SA (perception, comprehension, and projection; Chen et al., 2014). While the three levels of SAT-model transparency build on each other, the current data indicate that in certain task contexts there may not be a direct ordered link to the level of automation transparency. That is, operator access to what the automation projects as the decision outcome may not necessarily be a “higher” level of transparency; rather, projected outcome information may just be one type of transparency. This point notwithstanding, more research needs to be conducted concerning how Level 1+2+3 transparency can be designed to increase the accuracy of automation use across different types of task domains, without compromising decision time or workload. In order to advance theory and to guide future approaches in designing transparent automation, there is a continued need for studies to report findings such as ours that do not necessarily find that all forms of increased transparency benefit automation use, to avoid contributing to the “file drawer” problem—in which positive results are more likely to be published and subsequently bias meta-analyses by over-estimating effect sizes and distorting confidence intervals (Cumming, 2014). Future research could track operators’ eye movements (Bruder and Hasse, 2020) or gain verbal reports (e.g., think-aloud protocol) from operators to examine shifts in strategy use or attention allocation from the provision of transparency information, which may lead to a better

understanding of the psychological mechanisms underlying the use of transparent automation. Future research may also include co-variables such as mathematical and calculation/estimation skills to examine individual differences in the use of transparent automation (see Matthews et al., 2020).

Automation disuse and misuse rates are likely dependent on the relative perceived reliability of automation and could be further investigated through comparison with unaided human performance (e.g., Baber et al., 2019). We do not have data on how well participants performed the task manually without the help of the automation. However, given the training we provided and anecdotal reports, we approximate that participants would have been able to achieve 65%–70% manual accuracy. Thus it is slightly poorer than the Recommender reliability of 75% but not so poor that it would have been a useful strategy to follow the automation all of the time. In line with this, our data suggest that participants followed the Recommender on 70% of test trials (67%, 70%, and 73% of the test trials, for the Level 1, Level 1+2, and Level 1+2+3 conditions, respectively), thus, there were neither floor nor ceiling effects in automation use. Starke and Baber (2020) compared participants’ detection-task performance with and without a decision support system that highlighted relevant important information sources under different reliability conditions. They found that participants’ detection accuracy performance significantly improved when aided by the decision support system, however, this was only found in the one condition where the system was 100% reliable and when this reliability rate was made known to participants. Future research should investigate the effectiveness of the three levels of SAT-model transparency under conditions in which participants have also performed the task manually, and are told the automation reliability.

The simulated UV task used in the current study represents situations in which operators need to make decisions under time pressure on complex quantitative data. That said, we are aware of the potential problems in generalizing from novice participants to expert operators as there are undoubtedly differences in their cognitive skills and motivation. Future research is required to examine if expert UV operators are affected differently by our types of transparency designs. Clearly, however, more research is required to tease apart exactly what type of transparency information is beneficial for the operator and in what task contexts. As the automation transparency literature matures, we should be able to better shape and understand the fundamentals to designing an interface that supports transparency and decision-making in human-autonomy teaming.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Our co-author, Michael Skinner, sadly passed away during the preparation of this manuscript. He was a generous and highly respected colleague, had made significant intellectual contributions to this project, and will be greatly missed. We dedicate this paper to his memory.

Appendix. Mission Descriptions and Intel Messages

Training

Mission ID	Mission Map	Mission Text	Intel	Asset Name	Associated Plan	Correct Plan
4	Rural	Five people have not been accounted for in a bushfire (in southeastern region). Send a vehicle to search for these people.	A thunderstorm is expected to hit the area in about 12 h' time. A music festival will be conducted in the Eastern region next week. A road blockage has decreased the speed of UGV 2 by 3.	UGV 3 UGV 4 UGV 2 UAV 1	A B None None	A
6	Rural	A pilot made a mayday call at Sector X, before crashing and losing contact. Send a vehicle to find the location of the crash site.	UAV 2 is scheduled for regular maintenance tomorrow. UGV 4 is travelling through rough terrain, reducing its speed by 1. Heavy fog is reported in the Northern region.	UAV 1 UAV 2 UAV 3 UGV 4	A B None None	A
9	Rural	There has been a traffic accident in the south west suburbs. Send a vehicle to assess the situation.	Fog is expected in the Southeast mountains. UAV 4's updated GPS signals show that its position is actually 4 cells due West. Tailwind has increased UAV 3's speed by 4.	UGV 1 UAV 3 UAV 2 UAV 4	A B None None	B
14	Rural	Demolitions to create a new open-cut mine are due to commence. Monitor the area and report any vehicles or people detected.	Military base has tightened security levels. High octane fuel for UGV 1 has increased its fuel efficiency by 3. UAV 4 is due for a new paint job next week.	UGV 3 UGV 1 UAV 4 UAV 2	A B None None	B
17	Coastal	Illegal fishing has been reported North of the eastern islands. Patrol the area and intercept any vessels detected.	A pod of dolphins has been sighted off the North of the Eastern Islands. A problem with USV 1's engine has decreased its speed by 3. High octane for fuel UAV 2 has increased its fuel efficiency by 3.	USV 1 USV 3 UAV 2 USV 4	A B None None	A
22	Coastal	A large shipment of illegal drugs has been spotted in pirate-infested waters. Send a vehicle to identify and intercept the ship.	Worsened sea conditions have decreased USV 3's speed by 4. Storms last night caused tree damage in coastal areas. UAV 1 has been filled with high octane fuel, increasing its fuel efficiency by 2.	USV 3 USV 2 USV 4 UAV 1	A B None None	A
24	Coastal	Provide security and surveillance support for a fleet of cargo ships as they traverse pirate-infested waters to the north.	USV 4's updated GPS signals show that it is actually 4 cells due west. Whale watching season has begun. Engine problems for USV 2 has decreased its speed by 4.	USV 4 USV 1 USV 2 USV 3	A B None None	A
36	Urban	A large protest has erupted around Parliament House. Provide reconnaissance as events unfold.	A truck has become stuck under a bridge in the Northern Business District, causing a traffic blockage. UAV 1 is experiencing connectivity issues with the GPS, its current position is now 1 cell due north. A screw has come loose from the propeller of UGV 3, causing it to reduce its speed by 3.	UAV 2 UAV 1 UAV 4 UGV 3	A B None None	A
39	Urban	A peaceful demonstration is scheduled to commence soon. Send a vehicle to monitor this event.	A fire broke out at a home in the south-west region. UGV 3 is due for a service next week. UGV 3 has been given the wrong grade fuel, decreasing its fuel efficiency by 3.	UGV 3 UAV 2 UAV 4 UAV 1	A B None None	B
40	Urban	There is traffic congestion along the eastbound freeway. Send a vehicle to monitor the flow of traffic.	A four-vehicle crash is causing a traffic gridlock on the southern freeway. There is a hot air balloon show in the city. UGV 3 is caught in traffic, causing its speed to decrease by 3.	UAV 1 UAV 2 UAV 4 UGV 3	A B None None	A
44	Urban	An unidentified UAV is spotted flying close to several homes in the south eastern suburbs and hacking into home Wi-Fis. Send a vehicle to search, identify, and intercept.	Police department has received a report of a burglary in the southern suburb. UAV 1's GPS has been updated, its current position is now 2 cells due east. UAV 2's GPS has been updated, its current position is now 2 cells due south.	UAV 2 UAV 3 UAV 1 UGV 4	A B None None	A
45	Urban	An unidentified UAV is being used to spray paint and vandalize billboards off the Southern Freeway. Send a vehicle to intercept.	A man has been seriously injured in a traffic accident on the Southern Freeway. UAV 3 is due for a replacement of landing wheels. UAV 4 is having problems with its engine, causing it to slow down by 3.	UAV 3 UAV 4 UAV 2 UGV 1	A B None None	A

Test

Mission ID	Mission Map	Mission Text	Intel	Asset Name	Associated Plan	Correct Plan
1	Rural	A bushfire is burning out of control in eastern mountain ranges, send an air-vehicle to search for hikers trapped in the area.	Tailwind has increased UAV 2's speed by 3. There is a protest going on in the south bushland. Extra weight on UAV 1 has decreased its fuel efficiency by 3.	UAV 4 UAV 2 UAV 1 UAV 3	A B None None	B
2	Urban	Police are searching for a stolen car being driven around the northern business district. Send a vehicle to search the area.	Extra weight on UAV 3 has decreased its fuel efficiency by 4. A new fleet of helicopters has been ordered in. Headwind has reduced the speed of UAV 2 by 2.	UAV 3 UGV 1 UGV 4 UAV 2	A B None None	B
3	Rural	A pilot made a mayday call around the North East mountain ranges, before losing contact. Send a vehicle to check where and whether the aircraft has crashed and if there were any survivors.	Tailwind has increased UAV 3's speed by 1. Traffic is flowing smoothly on all roads. UAV 4 has been active for the last 300 h.	UGV 2 UAV 3 UAV 1 UAV 4	A B None None	A
5	Rural	An earthquake caused structural damage in some houses, send a vehicle to identify the damage.	An after shock is expected to hit the same area within the next days or so. UGV 1 has been filled with low quality fuel, decreasing its fuel efficiency by 3. UAV 3 requires repainting.	UGV 1 UAV 3 UGV 2 UAV 4	A B None None	A
7	Rural	A vehicle has been detected moving at high speed directly towards a Defense Base. Intercept the vehicle before it reaches the perimeter.	Tailwinds for UAV 4 has increased its speed by 3. Reduced weight on UGV 2 has increased its fuel efficiency by 2. There is a family fun day picnic in the south east mountains that may cause some ground traffic delays.	UGV 2 UGV 1 UAV 4 UGV 3	A B None None	A
8	Rural	Poachers have been reported to be hunting within protected national park. Send a vehicle to intercept them.	Improved traffic has increased UGV 3's speed by 4. Thunderstorms expected. Extra cargo on UAV 1 has decreased its fuel efficiency by 3.	UGV 4 UGV 3 UAV 1 UGV 2	A B None None	A
11	Rural	There is a fire in the southern bushlands. Send a vehicle to map hotspots and high-risk areas.	Due to headwinds, UAV 1's speed has decreased by 3. A truck has overturned in the Southeast mountains. UGV 3's wheels are ill-suited for snowy terrain, reducing its speed by 1.	UAV 1 UAV 2 UAV 4 UGV 3	A B None None	A
12	Rural	Reports of people illegally dumping rubbish on the beach. Send a vehicle to deter litterers.	Propeller damage to UAV 2 has decreased its speed by 3. UGV 4 will need new tyres after the next mission. UGV 1 will be used to transport politicians next week for the Commonwealth meeting.	UGV 4 UGV 1 UAV 2 UGV 3	A B None None	A
13	Rural	Heavy rains have caused rivers from the South Eastern mountains to flood. Send an air vehicle to monitor the flood waters.	UAV 4 will need a minor service after the next mission. A hot air balloon event is occurring in the North. Engine problems for UAV 1 has decreased its speed by 2.	UAV 4 UAV 2 UAV 1 UAV 3	A B None None	A
15	Coastal	A distress call is received from a boat North of Eastern Island due to engine failure. Send a vehicle to search for the exact location of the boat.	Sunny day means that many pleasure crafts are expected at sea. A jet ski has crashed into USV 1, causing its speed to reduce by 2. High octane fuel for UAV 3 has increased its fuel efficiency by 1.	UAV 2 UAV 3 USV 1 USV 4	A B None None	A
16	Coastal	A pilot made a mayday call off the west coast, before losing contact. Send a vehicle to check where and whether the aircraft has crashed and if there were any survivors.	Problems with USV 4's sensors have increased its search steps by 2. UAV 3 will be repainted within the next month. Ironman competitions are taking place in the south-east.	UAV 3 USV 4 USV 2 UAV 1	A B None None	A
18	Coastal	A pilot made a mayday call above the pirate-infested water, before crashing and losing contact. Send a vehicle to check the exact location of the aircraft and if there were any survivors.	Due to a holding pattern, UAV 4's speed has been reduced by 1. Impact with a bird has increased the search steps of UAV 4 by 2. There's a sailing vessel spotted at Cell B2.	USV 1 USV 2 USV 3 UAV 4	A B None None	A
19	Coastal	Unidentified cargo ship is planning to dock on the island's south coast; send a vehicle to inspect and identify the ship.	Hot day has produced large beach crowds. A blocked hose has reduced USV 1's fuel efficiency by 1.	UAV 2 USV 1 USV 4 USV 3	A B None None	A

(continued on next page)

(continued)

Mission ID	Mission Map	Mission Text	Intel	Asset Name	Associated Plan	Correct Plan
20	Coastal	A suspicious boat is operating aimlessly around the Eastern Islands. Provide surveillance for a potential attack.	UAV 2 will be featured in an exhibition next month. USV 1's search steps have increased by 2 due to poor sensors. Weather forecast for tomorrow is calm. Updated GPS coordinates show that UAV 3 is 3 cells due East.	USV 4 UAV 3 USV 1 UAV 2	A B None None	B
21	Coastal	A distress call has been received from a ship off the West Coast. Send a vehicle to monitor and assist rescue operations.	Traffic is congested on the roads to the beach. Westerly winds have increased the speed of USV 3 by 3. GPS signals for USV 4 has been updated. It is now 2 cells due South.	USV 1 USV 3 USV 2 USV 4	A B None None	A
23	Coastal	The King's aircraft has lost communications off the Western Beaches. Find it as soon as possible.	UAV 4's GPS has been updated, its current position is now 2 cells due West. The commander has just received a long service award. UAV 1's updated GPS shows that it is actually positioned 2 cells due West.	UAV 1 UAV 2 UAV 4 USV 3	A B None None	A
25	Coastal	Empty rocket boosters have detached from the country's first spaceship and are expected to land off the north coast. Monitor the area to track its exact landing location.	A ship has been rescued in pirate-infested waters. USV 4 will need an oil change within the next week. USV 2 has been fitted with improved sensors, decreasing search steps by 3.	UAV 3 USV 2 USV 4 UAV 1	A B None None	B
26	Coastal	Swimming marathon is on between the eastern islands, and a shark has been sighted nearby. Monitor the area for the duration of the swimming event.	USV 3's speed has been reduced by 3 due to unexpected choppy conditions. UAV 2 has been injected with high octane fuel, causing its fuel efficiency to increase by 3. UAV 1 has been scheduled for an air force exhibition next month.	UAV 1 UAV 2 UAV 4 USV 3	A B None None	A
27	Coastal	A giant oil rig to the West is leaking oil into the ocean. Monitor the area to track the oil as it spreads.	Updated GPS signals show that USV 1 is actually 4 cells due North. UAV 3 is experiencing headwind, causing its speed to decrease by 4. New shipping lanes will open up in this sector in the next two weeks.	UAV 4 USV 1 USV 2 UAV 3	A B None None	A
28	Coastal	A ferry is reported sinking after colliding with a cargo ship. Twenty people remained unaccountable in waters near the Western Beaches. Send a vehicle to find these people.	USV 1 has been scheduled for a military magazine feature next week. USV 2's servicing vendor has been changed to a different company. USV 4's GPS has been updated, its current position is actually 4 cells due east.	USV 4 USV 2 USV 1 USV 3	A B None None	B
29	Urban	A suspicious package has been found in the football stadium. Send a vehicle to inspect before the game starts and crowd arrives.	Building collapses in the southern region after catching fire. UAV 1's experiencing a tailwind, causing its speed to increase by 3. UGV 4 has a bad GPS signal causing it to reduce its coverage to 1.	UAV 3 UAV 1 UGV 4 UAV 2	A B None None	A
30	Urban	Buildings have collapsed due to an earthquake. Search for survivors.	UGV 2 has been filled with high octane fuel increasing its speed by 4. West beaches are closed due to heavy storm. UAV 3 has trouble accelerating, its speed is dropping by half.	UGV 4 UGV 2 UGV 1 UAV 3	A B None None	B
31	Urban	An armed fugitive is hiding in abandoned building. Send a vehicle to provide surveillance for police.	UGV 2 has been fitted with new turbo engines, increasing its speed by 4. More than 30 people injured as protesters storm the Parliament near the Northern Business District. UGV 1 fuel pump is malfunctioning, causing its fuel efficiency to reduce by 1.	UAV 3 UGV 1 UAV 4 UGV 2	A B None None	A
32	Urban	A group of teenagers have been seen breaking into and vandalizing warehouses. Send a vehicle to the area to confirm this intel.	Plan is set to build major new rail lines in the north within the next two years. UAV 1's GPS has been updated, its current position is now 2 cells due east. Rougher than expected terrain has reduced the fuel efficiency of UGV 2 by 4.	UGV 2 UGV 3 UAV 1 UAV 4	A B None None	A
33	Urban	A suspicious boat is about to dock on the beach near the prison. Send an air vehicle to provide surveillance of the area.	Confirmed crocodile sighting in the West River. UAV 2 is sputtering and lower power, causing its speed to reduce by 2. UAV 3 has wing damage, causing it to decrease its speed by 4.	UAV 3 UAV 4 UAV 2 UAV 1	A B None None	B

(continued on next page)

(continued)

Mission ID	Mission Map	Mission Text	Intel	Asset Name	Associated Plan	Correct Plan
34	Urban	An unidentified aircraft has been detected flying around a maximum security prison. Search for and intercept the aircraft as soon as possible.	Crash involving two cars causing chaos in the Southern Freeway. UAV 4 is experiencing engine problems, causing its speed to decrease by 2. UAV 1 is scheduled for repainting next week.	UAV 1 UAV 3 UAV 2 UAV 4	A B None None	A
35	Urban	A high-value prisoner is being transferred between prisons. Provide surveillance over their route as they leave the city.	UAV 3 will need an oil change within the next week. Birdwatching tourists are being rescued from mud in the lower south east region. UGV 2 encountered a headwind, causing its speed to decrease by 1.	UGV 1 UGV 2 UAV 3 UGV 4	A B None None	A
37	Urban	Escort Air Force One as it returns to the US. Maintain surveillance of the surrounding airspace.	Passengers are stuck in the crowded Central Airport due to flight delays. UAV 4 is scheduled for maintenance tomorrow. Suspected damaged tyres on UAV 1.	UAV 4 UAV 3 UAV 1 UAV 2	A B None None	A
38	Urban	Violence has erupted at the football stadium. Send a vehicle to provide support to police.	Traffic delays are expected near the football stadium. UAV 1 encountered a headwind, causing its speed to decrease by 3. UAV 4 is experiencing connectivity issues with the GPS, its current position is now 3 cells due south.	UAV 1 UAV 3 UAV 2 UAV 4	A B None None	A
41	Urban	The International Airshow is on. Maintain surveillance of the surrounding airspace to ensure non-authorised aircraft do not enter.	UAV 4 has been filled with poor grade fuel, reducing its fuel efficiency by 3. Flood waters are blocking traffic in the north eastern region. Excessive rain in the north eastern region, causing UGV 3 to reduce its speed by 1.	UAV 4 UAV 1 UGV 2 UGV 3	A B None None	A
42	Coastal	Suspicious submerged object has been reported in the harbour. Send a vehicle to identify.	A shark warning has been re-issued around the east beaches. USV 1 has been detected by pirates, resulting in its speed increasing by 2. UAV 4's updated GPS signals show that it is actually 4 cells due West.	UAV 4 UAV 2 USV 1 USV 3	A B None None	B
43	Coastal	A cruise ship has been quarantined due to a virus outbreak as it docks at one of the northern islands. Send a vehicle to deliver medical supplies.	Western beaches have been closed due to extreme weather warning. UAV 1's new cruising altitude has increased, allowing the speed to increase by 5. USV 4 is showing signs of mechanical faults, decreasing its fuel efficiency by 2.	UAV 3 UAV 1 USV 4 UAV 2	A B None None	A
46	Rural	Crops have been damaged on farms in the northern region. Send a vehicle to assess the extent of the damage.	There is heavy traffic in the central district. A boy is reported missing in the North Park. UAV 4 has a bad GPS signal causing it to reduce its coverage to 1.	UAV 1 UAV 3 UAV 2 UAV 4	A B None None	A
47	Rural	The southern region is holding its annual health fair. Send a vehicle to deliver medicine and other medical supplies.	Riots are occurring in the south west region. UAV 3 has temporarily lost its radio communication. UAV 2 is flying against the wind, causing its speed to decrease by 2.	UAV 1 UAV 2 UAV 3 UAV 4	A B None None	A
48	Rural	A report is received about a drone spying on homes in the southern region. Send a vehicle to find and track it.	A new theme park is opening in the mid-west. The North Farms community has received an upgrade to its facilities. UAV 2 is leaking fuel, causing its fuel efficiency to reduce by 3.	UAV 2 UGV 1 UAV 4 UGV 3	A B None None	B

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